



Ramakrishna Mission Vivekananda Educational and Research Institute

PO Belur Math, Howrah, West Bengal 711 202

School of Mathematical Sciences

Department of Computer Science

MSc BDA : Batch 2023-25, Semester I, Mid Exam

DA102: Basic Statistics

Dr. Sudipta Das

Student Name (in block letters):

Date: 19 September 2023

Student Roll No:

Max Marks: 60

Signature:

Time: 2hrs

Answers must be properly justified to deserve full credits.

1. (12 points)

Choose the correct answer.

- (a) (1 point) Students are asked to give a feedback about usefulness of the statistics course on a scale of 1 to 10. This type of variable is
1. Nominal in nature; 2. Ordinal in nature; 3. Discrete in nature; 4. Continuous in nature.
- (b) (1 point) Which of these represent qualitative data
1. Height of a student; 2. Liking or disliking of (500) persons of a product; 3. Income of a government servant in a city; 4. Yield from a wheat plot.
- (c) (1 point) In a Pie calculate the angles for each sectors by the following formula
1. $\frac{\text{componentPart}}{\text{Total}} \times 100$; 2. $\frac{\text{componentPart}}{\text{Total}} \times \pi$; 3. $\frac{\text{Total}}{\text{componentPart}} \times 360^\circ$; 4. $\frac{\text{componentPart}}{\text{Total}} \times 360^\circ$.
- (d) (1 point) If each of a set of observation of a variable is multiplied by a constant (non-zero) value, the variance of the resultant variable
1. is unaltered; 2. increases; 3. decreases; 4. is altered.
- (e) (1 point) A Histogram contains a set of
1. Adjacent rectangles; 2. Non Adjacent Rectangles; 3. Adjacent squares; 4. Adjacent triangles.
- (f) (1 point) When $X = \{x_i\}$ and $Y = \{y_i\}$ are two variables ($i = 1, 2, \dots, n$) with G.M.'s G_1 and G_2 , respectively then the geometric mean of $Z = \frac{X}{Y} = \left\{ \frac{x_i}{y_i} \right\}$ is
1. $\frac{G_1}{G_2}$; 2. $\text{antilog} \left(\frac{G_1}{G_2} \right)$; 3. $n(\log G_1 - \log G_2)$; 4. $\text{antilog} \left(\frac{\log G_1 - \log G_2}{2n} \right)$.
- (g) (1 point) A distribution in which the observations are concentrated at one end of the distribution is called a
1. Symmetric distribution; 2. Normal distribution; 3. Skewed distribution; 4. Uniform distribution.
- (h) (1 point) Which of the following is not an unit-less measure
1. Bowley's coefficient of skewness; 2. Gini's coefficient; 3. quartile deviation; 4. coefficient of quartile deviation.
- (i) (1 point) Histogram can be drawn only for
1. Discrete frequency distribution; 2. Cumulative frequency distribution 3. Continuous frequency distribution; 4. Relative frequency distribution.

- (j) (1 point) In a histogram the area of each rectangle is proportional to
1. the class mark of the corresponding class interval;
 2. the class size of the corresponding class interval;
 3. frequency of the corresponding class interval;
 4. cumulative frequency of the corresponding class interval.
- (k) (1 point) Census reports used as a source of data is
1. Primary source;
 2. Secondary source;
 3. Organized data;
 4. None
- (l) (1 point) Frequency polygon can be drawn with the help of
1. Histogram;
 2. Circle;
 3. Percentage.

2. (14 points)

In two sets of variables x and y with 40 observations each, the following data were observed:

$$\bar{x} = 10, \sigma_x = 3, \bar{y} = 5.5, \sigma_y = 2 \text{ and } \text{cor}(x, y) = 0.28.$$

But on subsequent verification it was found that one value of $x (= 10)$ and one value of $y (= 5.5)$ were inaccurate and hence weeded out. With the remaining 49 pairs of values, how is the original value of $\text{cor}(x, y)$ affected?

3. (14 points)

The following table gives the distribution of the masses (in grams) of 50 salmon from a single young cohort.

Weight	155-164	165-174	175-184	185-194	195-204
Frequency	8	11	18	9	4

Sketch a histogram of the data. Comment on the skewness of this data distribution by comparing the mean and the median.

4. (20 points)

The following are midterm and final examination test scores for 10 students from a calculus class, where x denotes the midterm score and y denotes the final score for each student.

x	68	87	75	91	82	77	86	84	76	79
y	74	83	80	93	86	79	97	95	89	92

- (a) (6 points) Find the value of Spearman's rho between x and y .
- (b) (6 points) Find the value of Kendall's tau between x and y .
- (c) (8 points) Calculate the least-squares regression line of y on x for these data.

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This exam has total 4 questions, for a total of 60 points and 0 bonus points.
